Suggested problems: 1-49 odd

Problem 1: Describe the curve that is parameterized by

$$x = t \quad y = t^2.$$

The parabola $y = x^2$.

Problem 2: Describe the curve that is parameterized by

$$x = -\cos(t) \quad y = \sin(t).$$

The unit circle, trace clockwise.

Problem 3: Describe the curve that is parameterized by

$$x = \cos t \quad y = \sin t \quad z = t.$$

A helix wrapping around the z-axis of radius 1.

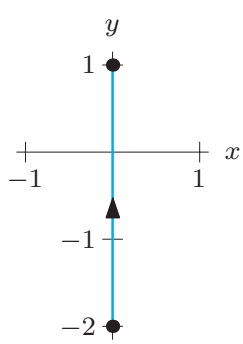
Problem 4: Describe the curve that is parameterized by

$$x = 2\sin t \quad y = 2\cos t \quad z = -t.$$

A helix wrapping around the z-axis of radius 2.

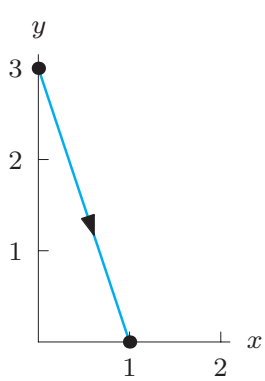
Problem 5: Find parameterizations for the following line segments.

(a)



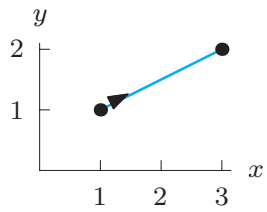
$x = 0$, $y = -2 + 3t$

(b)



$x = t$, $y = 3 - 3t$

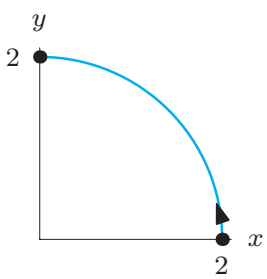
(c)



$x = 1 + 2t,\ y = 1 + t$

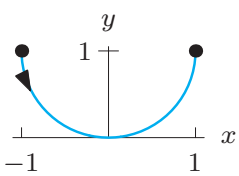
Problem 6: Find parameterizations for the following curves.

(a)



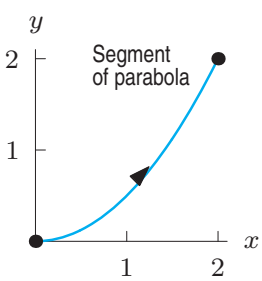
$x = 2\cos t,\ y = 2\sin t$ with t in $[0, \pi/2]$

(b)



$x = \cos t,\ y = \sin t + 1$ with t in $[\pi, 2\pi]$

(c)



$x = t,\ y = t^2/2$ with t in $[0, 2]$

Problem 7: Find a parameterization for the following curves.

(a) A circle of radius 3 centered on the z-axis and lying in the plane $z = 5$.

$$x = 3\cos t, \quad y = 3\sin t, \quad z = 5$$

(b) The circle of radius 2 in the xy-plane centered at the origin, clockwise.

$$x = 2\sin t, \quad y = 2\cos t, \quad z = 0$$

(c) The circle of radius 2 in the xz-plane centered at the origin.

$$x = 2\cos t, \quad y = 0, \quad z = 2\sin t$$

(d) The curve in which the plane $z = 2$ cuts the surface $z = \sqrt{x^2 + y^2}$.

$$x = 2\cos t, \quad y = 2\sin t, \quad z = 2$$

Problem 8: Find a parameterization of a line from $P = (2, 5)$ to $Q = (12, 9)$ so that P corresponds to $t = 1$ and Q to $t = 5$.

$$x = 2 + \frac{10}{4}(t - 1), \quad y = 4 + t$$